

Making Sense of Traffic

Building an AI-Powered Vehicle Tracker & Counter

The Traffic Problem

- ✓ Traffic is getting worse, and counting cars manually is incredibly slow and tedious.
- ✓ City planners and map apps need real-time data to actually fix congestion.
- ✓ We need a system that can watch a road and automatically count the vehicles.
- ✓ It shouldn't just count moving shapes—it needs to know if it's a car, truck, or bike!

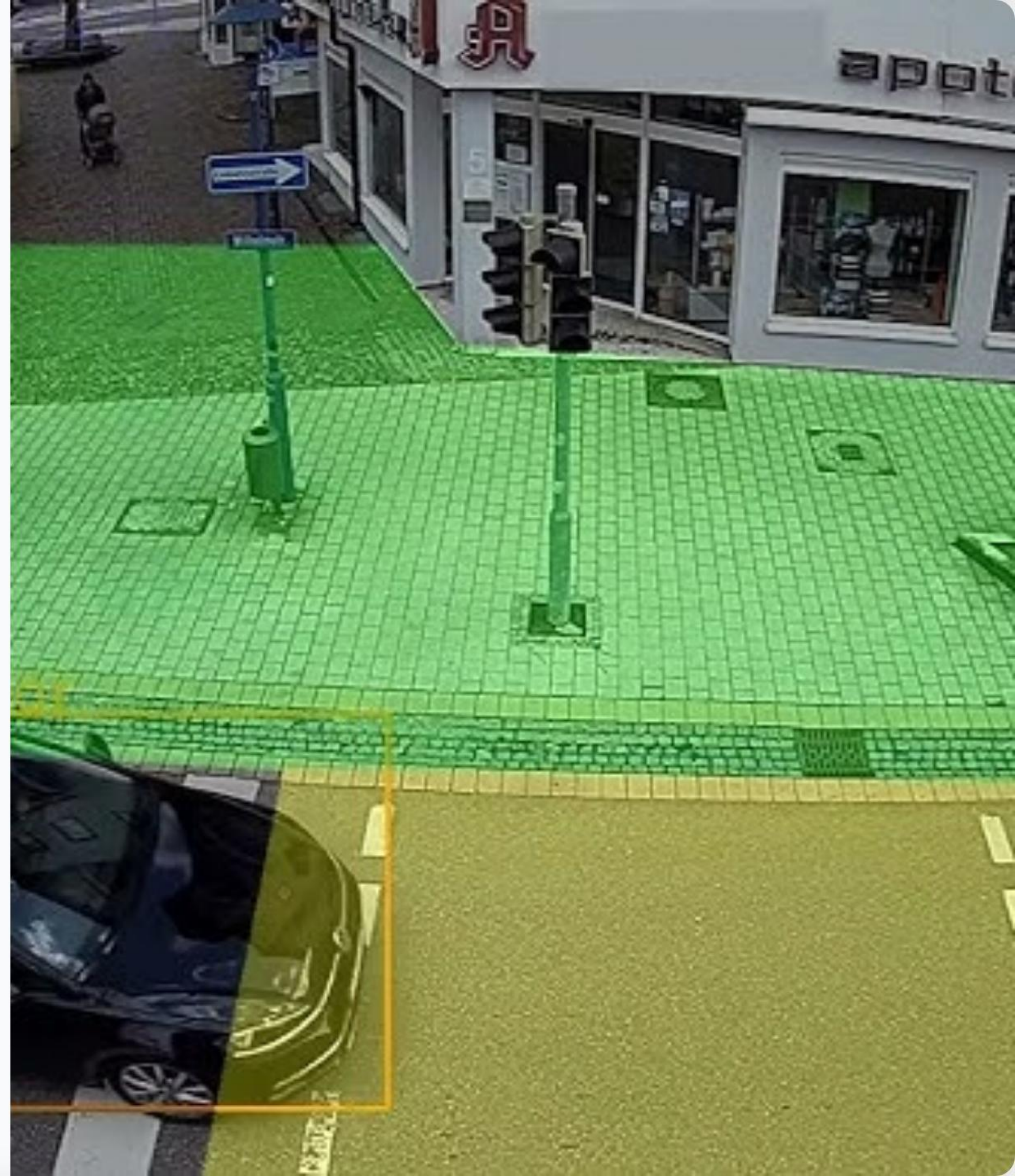


The AI Solution

A Tireless Digital Cop

Instead of humans, we use a simple Python script and a camera feed. This acts as an automated traffic monitor that never gets tired.

It processes video in real-time, instantly spotting vehicles, drawing boxes around them, and keeping a live tally on the screen.



Meet the Brain: YOLO11



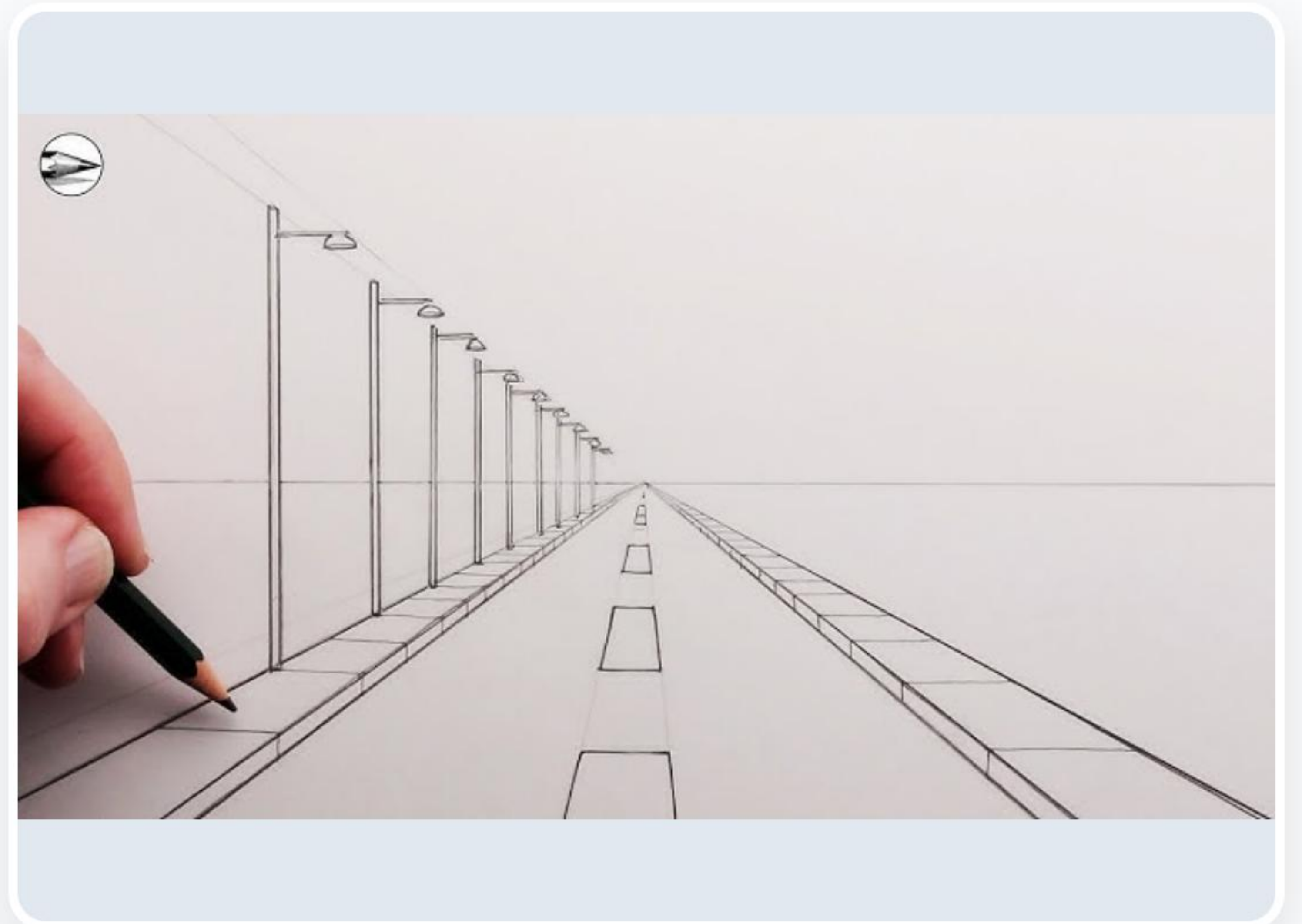
You Only Look Once

We didn't have to teach our program what a car looks like. We gave it a pre-trained "brain" called YOLO11.

Because it has already studied millions of images, it natively recognizes over 80 everyday items right out of the box. We just told it to look for vehicles!

The "Tripwire" Concept

- ✓ How do we count? We draw a virtual red line across the road in our video.
- ✓ The AI calculates the exact center point of every vehicle it sees.
- ✓ When a vehicle's center crosses our red line, it adds to the tally.
- ✓ This ensures we count cars accurately, regardless of their speed!



Giving AI a Memory



1. Detect

First, the AI spots a vehicle in the video frame and figures out its exact coordinates and bounding box.



2. Remember

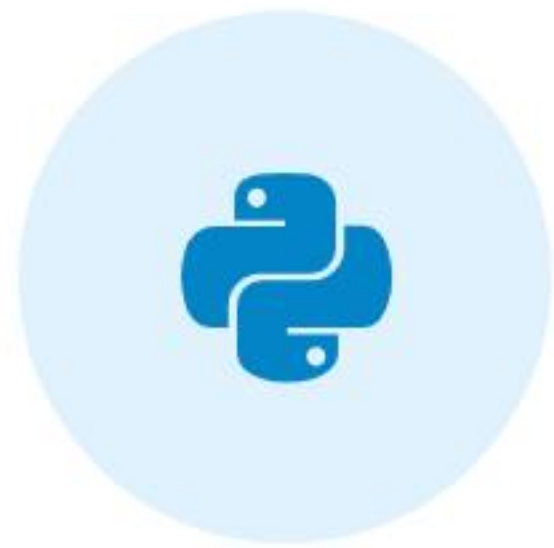
By enabling "tracking", the AI assigns a unique ID (like #123) to the car so it doesn't forget it in the next frame.



3. Count Once

If ID #123 crosses the line, we count it and save the ID. If we see #123 again, we safely ignore it!

What's Under the Hood?



Python

The glue that holds it all together. It's the language we use to write the rules and logic for the tracking system.



OpenCV

The "eyes" of the project. It handles opening the video file, drawing our red line, and displaying numbers on screen.



Ultralytics

The powerful library that lets us easily plug the state-of-the-art YOLO11 AI engine right into our code.

What's Next?

Speed tracking, smart traffic lights, and beyond.



Any Questions?

Image Sources



https://static.vecteezy.com/system/resources/previews/071/533/724/large_2x/dense-traffic-with-many-cars-and-motorbikes-on-a-city-road-during-daytime-creating-a-long-queue-line-photo.jpeg

Source: www.vecteezy.com



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Source: www.youtube.com



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Source: www.vecteezy.com



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